

TASK 1 — PROJECT SELECTION & PROBLEM STATEMENT

1.1 Project Title

Student Performance Prediction Using Machine Learning

1.2 Introduction

I selected **Student Performance Prediction** as my final AI/ML project.

Academic performance is an important factor in determining a student's educational progress. Many factors such as study time, previous failures, absences, family education, social activities, and health can influence a student's academic results.

In this project, I use machine learning to analyze these factors and predict whether a student is likely to **Pass or Fail**.

1.3 Problem Statement

Educational institutions have large amounts of student-related data, but identifying students who may perform poorly can be difficult using traditional methods.

The problem addressed in this project is:

Can machine learning be used to predict whether a student will pass or fail based on academic, demographic, and behavioral factors?

The project develops a machine learning classification model that learns patterns from historical student data and predicts the performance category of a student.

1.4 Objective

The objectives of my project are:

1. To collect a suitable student performance dataset.
2. To understand the structure and characteristics of the dataset.
3. To clean and preprocess the data.
4. To identify important features related to student performance.

5. To perform Exploratory Data Analysis (EDA).
 6. To visualize relationships between student characteristics and performance.
 7. To develop a machine learning classification model.
 8. To train the model using historical student data.
 9. To predict student performance.
 10. To evaluate the model using appropriate classification metrics.
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1.5 Why AI/ML Is Suitable

I selected machine learning because student performance depends on multiple variables.

For example:

- Study time can affect performance.
- Previous failures may indicate academic difficulties.
- Absences can affect learning.
- Family education may have an influence.
- Social and health-related factors can also be relevant.

Instead of manually analyzing every factor, machine learning can identify patterns from historical data and use those patterns to make predictions for new students.

1.6 Expected Outcome

The expected outcome is a machine learning model that can classify students into:

Pass or Fail

based on selected student characteristics.

The project should also help identify which features have a stronger relationship with student performance.

1.7 Real-World Application

This system could be useful for:

- Schools
- Colleges
- Universities
- Teachers
- Academic counselors

- Student support departments

For example, an institution could use such a system as an **early-warning tool** to identify students who might need additional academic support.

1.8 Expected Output

At the end of Task 1, my notebook/report would contain:

Project Title:

Student Performance Prediction Using Machine Learning

Problem:

Predict whether a student will pass or fail based on academic, demographic, and behavioral information.

Input:

Student-related features

Output:

Pass / Fail

Machine Learning Type:

Classification

Selected Algorithm:

Random Forest Classifier
